

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

TAE-I (Session 2026-27) Winter Term 2026

TAE-I	Subject: Integrating AI with DBMS	
Department: AI	Semester: V	Section: A/B/C
Unit. No: 1,2	Date of Assignment given: 21-07-2026	
Name of Faculty: Prof. Binaya Kumar Suna / Prof. Prachi Durge	Date of Assignment Submission: 27-07-2026	

Dataset Information (archive.zip Present in TAE-I Folder) :

The dataset is structured for a retail and customer analytics system, comprising six interconnected tables that capture comprehensive customer, operational, and marketing data (Dataset Schema information in mysql_schema_specification.pdf in TAE-I Folder):

- **customers:** Stores demographic information, physical addresses, contact details, and account registration dates.
- **transactions:** Tracks purchase history, including products bought, quantities, unit prices, store locations, and payment methods.
- **interactions:** Captures digital behavior and activity logs, such as product views, cart additions, session durations, and channel usage.
- **campaigns:** Details marketing and advertising campaign performance metrics, including budgets, impressions, clicks, and conversions.
- **customer_reviews_complete:** Contains post-purchase product feedback, star ratings, and written review narratives submitted by customers.
- **support_tickets:** Logs customer service inquiries, issue categories, priority levels, resolution statuses, and satisfaction scores.

Steps for Submitting Assignment Via GitHub

Step 1: Clone the repository git clone <https://github.com/binayasuna-afk/IAI-DBMS.git>

Step 2: Move to the folder cd IAI-DBMS

Step 3: Create a new branch (Branch name must be your Roll No; Sections D, E, F use section_name-roll_no)
git checkout -b section_name-roll_no

Step 4: Complete your work and push git add . git commit -m "Assignment Submission" git push -u origin section_name-roll_no

Step 5: Create a Pull Request Go to <https://github.com/binayasuna-afk/IAI-DBMS>, click "Compare & pull request", and submit for review.

Your assignment solution should be submitted in 3 sql script namely Q1.sql, Q2.sql and Q3.sql.

G H Raisonni College of Engineering & Management

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Q. No.	Questions	CO Mapped
Q1	Create all tables based on the dataset and load the data to your local mysql database.	CO1
Q2	a) (Easy): Basic Filtering & Ordering: Write a MySQL query to retrieve the full_name, email, city, and preferred_channel of all customers residing in either 'California' or 'Texas' who registered after January 1st, 2023. Sort the results alphabetically by full_name. b) (Intermediate): Multi-Table Aggregation with Conditional Filtering Write a MySQL query to identify high-value customers who have spent a total of over \$1,000 across at least 3 separate transactions in the Online store. Display the customer's customer_id, full_name, email, the total amount spent (aliased as total_spent), and the total number of transactions. Format total_spent to 2 decimal places and order the output by total spending in descending order.	CO1
Q3	a) (Intermediate): Support Metrics & Category Filtering Write a MySQL query to evaluate urgent customer support tickets. Calculate the average resolution time in hours (resolution_time_hours) and average customer satisfaction score (customer_satisfaction_score) for each issue_category where the ticket priority is 'High' and the ticket status is 'Closed'. Round both calculated averages to 2 decimal places, and only include issue categories that have processed more than 5 high-priority tickets. b) (Very Hard): Multi-Table CTE / Cross-Channel Behavioral Analysis Marketing needs to analyze customer activity across communication channels. Write a single MySQL query using Common Table Expressions (CTEs) or subqueries to calculate the following metrics grouped by preferred_channel (from the customers table): - Total registered customers - Total count of 'Add to Cart' interactions (from the interactions table) - Total revenue generated (from the transactions table) - Total count of support tickets raised (from the support_tickets table) Note: Ensure all preferred communication channels are included in the output even if they have zero transactions or support tickets. Sort the results by total revenue in descending order.	CO2

Prof. Binaya Kumar Suna / Prof. Prachi Durge

Subject Faculty

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